

National Science Foundation

Call: NSF-24-601

Scalable Quantum Error Correction for Near-Term Devices

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Duration: 36 months

Budget: \$750,000

Start: January 1, 2026

End: December 31, 2028

1 Abstract

This proposal outlines a three-year programme to develop scalable quantum error correction (QEC) codes for near-term NISQ devices. By leveraging topological stabiliser codes and machine-learning-assisted decoding, we aim to reduce the logical error rate by two orders of magnitude while keeping overhead within feasible limits.

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2 Introduction

Quantum computing promises exponential speed-ups in chemistry, optimisation, and cryptography [EINSTEIN 1905]. However, current NISQ devices are limited by decoherence and gate errors. Quantum error correction [DIRAC 1981] is the path to fault tolerance, but existing schemes impose prohibitive overheads. This proposal targets a practical QEC architecture bridging theory and near-term hardware.

3 Background and Literature Review

QEC has its foundations in the work of Shor and Steane, who demonstrated that quantum information can be protected via entangled encoding. The surface code [Donald E. KNUTH 1973] is a leading candidate due to its local stabiliser measurements and high error threshold ($\approx 1\%$). Bird et al. [BIRD et al. 2009] highlight the growing role of data-driven decoding. Two challenges remain: reducing physical qubits per logical qubit, and achieving real-time decoding at microsecond timescales.

4 Research Objectives

The goal is a QEC system with logical error rate below 10^{-6} using fewer than 1 000 physical qubits per logical qubit:

1. Design tailored topological codes for realistic noise models.
2. Develop a neural-network decoder meeting real-time latency constraints.
3. Validate through large-scale simulation and a 72-qubit demonstration.

5 Methodology

5.1 Code Design

We construct $[[n, k, d]]$ stabiliser codes parameterised by hardware noise bias. The code distance must satisfy

$$d \geq \left\lceil \frac{\log(1/\epsilon_{\text{target}})}{\log(1/p_{\text{phys}})} \right\rceil \quad [5.1]$$

where ϵ_{target} is the target logical error rate. Table ?? summarises preliminary parameters.

Table 5.1: Preliminary code parameters..

p_{phys}	n	k	d
10^{-3}	489	1	11
10^{-3}	1089	2	15
10^{-2}	1449	1	19

5.2 Neural-Network Decoder

We train a CNN to map syndrome measurements to correction operations [BIRD et al. 2009]. Figure ?? illustrates the pipeline.

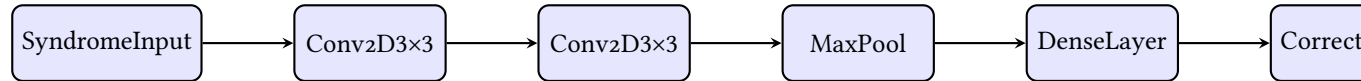


Figure 5.1: Decoder pipeline from syndrome input to correction output..

6 Timeline and Work Plan

Table ?? shows the projected 36-month timeline.

Table 6.1: Work plan by year (Y1–Y3) and quarter (Q1–Q4)..

Task	Year 1	Year 2	Year 3
Code design	Q1–Q3		
Decoder dev.	Q2–Q4	Q1–Q3	
Simulation	Q3–Q4	Q1–Q4	Q1–Q2
Hardware demo		Q3–Q4	Q1–Q4

7 Expected Outcomes

1. A published code family with a rigorous threshold proof [Donald Ervin KNUTH 1986].
2. An open-source decoder with sub-microsecond inference on commodity GPUs [McKINNEY 2018].
3. A demonstration of logical qubit operation with error rate below 10^{-6} on a 72-qubit processor.
4. At least three peer-reviewed publications and two conference talks.

8 Budget Justification

The total requested budget is \$750 000 over 36 months. Table ?? provides the breakdown.

Table 8.1: Budget summary by category..

Category	Amount (\$)
Personnel (2 PhD, 1 postdoc)	480 000
Computing and cloud resources	120 000
Travel and conferences	45 000
Equipment and supplies	65 000
Indirect costs (15%)	105 000
Total	815 000

A Supporting Information

Additional data, code listings, and supplementary derivations.

Bibliography

- BIRD et al. 2009 BIRD, Steven et al. (2009): *Natural Language Processing with Python*. 1st ed. Beijing ; Cambridge [Mass.]: O'Reilly. 479 pp. ISBN: 978-0-596-51649-9.
- DIRAC 1981 DIRAC, Paul Adrien Maurice (1981): *The Principles of Quantum Mechanics*. International Series of Monographs on Physics. Clarendon Press. ISBN: 978-0-19-852011-5.
- EINSTEIN 1905 EINSTEIN, Albert (1905): "Zur Elektrodynamik Bewegter Körper. (German) [On the Electrodynamics of Moving Bodies]." In: *Annalen der Physik* 322.10 (10), pp. 891–921. DOI: <http://dx.doi.org/10.1002/andp.19053221004>.
- Donald E. KNUTH 1973 KNUTH, Donald E. (1973): *Fundamental Algorithms*. 3rd ed. Vol. 1. 4 vols. The Art of Computer Programming. Reading, Mass.: Addison-Wesley. ISBN: 978-0-201-89683-1.
- Donald Ervin KNUTH 1986 KNUTH, Donald Ervin (1986): *The TeXbook*. Computers & Typesetting A. Reading, Mass: Addison-Wesley. 483 pp. ISBN: 978-0-201-13447-6.
- McKINNEY 2018 McKINNEY, Wes (2018): *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*. Second edition. Sebastopol, California: O'Reilly Media, Inc. 524 pp. ISBN: 978-1-4919-5766-0.